1 Vypočtěte:

1) $\sqrt{5}+3\sqrt{3}-(2\sqrt{5}-3\sqrt{5}-\sqrt{3})$;
2) $5(\sqrt{2}+\sqrt{3})-4(\sqrt{3}-\sqrt{2})-3(3\sqrt{2}-\sqrt{3})$;
3) $\frac{2}{3}\sqrt{6}-\frac{1}{5}\sqrt{5}-\frac{3}{4}(\sqrt{6}-\sqrt{5})-\frac{1}{12}(11\sqrt{5}-\sqrt{6})$;
4) $8\sqrt{a}+5\sqrt{x}-7\sqrt{a}+4\sqrt{a}-6\sqrt{x}-3\sqrt{a}+\sqrt{x}$;
5) $4(\sqrt{a}+\sqrt{b})-2(\sqrt{a}-\sqrt{b})-6(3\sqrt{a}+\sqrt{b})+16\sqrt{a}$;
6) $a(\sqrt{a}+\sqrt{b})-\sqrt{a}(a-b)-a\sqrt{b}+b\sqrt{a}$.

[**1)** $4\sqrt{3}+2\sqrt{5}$; **2)** $4\sqrt{3}$; **3)** $-\frac{11}{30}\sqrt{5}$; **4)** $2\sqrt{a}$, $a\geq 0\ \wedge\ x\geq 0$; **5)** 0, $a\geq 0\ \wedge\ b\geq 0$; **6)** $2b\sqrt{a}$, $a\geq 0\ \wedge\ b\geq 0$.]

2 Vypočtěte:

1) $\sqrt{12}\cdot\sqrt{3}$;
2) $(\sqrt[3]{2}+\sqrt[3]{9})\sqrt[3]{3}$;
3) $\sqrt{10}\cdot\sqrt{12}\cdot\sqrt{30}$;
4) $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})$;
5) $\sqrt[3]{50}\cdot\sqrt[3]{20}$;
6) $\sqrt{2a}\cdot\sqrt{2b}\cdot\sqrt{ab}$, $a\geq 0\ \wedge\ b\geq 0$;
7) $\sqrt[3]{0{,}216}\cdot\sqrt[3]{125}$;
8) $\sqrt[3]{2a}\cdot\sqrt[3]{3b}\cdot\sqrt[3]{a}\cdot\sqrt[3]{ab}$, $a\geq 0\ \wedge\ b\geq 0$;
9) $(\sqrt{2}+\sqrt{3})\sqrt{2}$;
10) $(\sqrt[3]{a}+1)(\sqrt[3]{a^2}-\sqrt[3]{a}+1)$, $a\geq 0$.

[**1)** 6; **2)** $3+\sqrt[3]{6}$; **3)** 60; **4)** 1; **5)** 10; **6)** $2ab$; **7)** 3; **8)** $a\sqrt[3]{6b^2}$; **9)** $2+\sqrt{6}$; **10)** $a+1$.]

3 Odmocněte částečně:

1) $\sqrt{12}$, $\sqrt{50}$, $\sqrt{72}$, $\sqrt{240}$, $\sqrt{315}$;
2) $\sqrt[3]{48}$, $\sqrt[3]{250}$, $\sqrt[3]{128}$, $\sqrt[3]{500}$, $\sqrt[4]{162}$;
3) $\sqrt{9a^3b}$, $\sqrt{4a^5b^3}$, $a\geq 0$, $b\geq 0$;
4) $\sqrt{27a^9b^{11}c^{21}}$, $\sqrt[3]{8a^7b^8}$, $\sqrt[3]{ab^{10}}$, $\sqrt[6]{a^5b^6}$, $a\geq 0\ \wedge\ b\geq 0\ \wedge\ c\geq 0$.

[**1)** $2\sqrt{3}$, $5\sqrt{2}$, $6\sqrt{2}$, $4\sqrt{15}$, $3\sqrt{35}$; **2)** $2\sqrt[3]{6}$, $5\sqrt[3]{2}$, $4\sqrt[3]{2}$, $5\sqrt[3]{4}$, $3\sqrt[4]{2}$; **3)** $3a\sqrt{ab}$, $2a^2b\sqrt{ab}$; **4)** $3a^4b^5c^{10}\sqrt{3abc}$, $2a^2b^2\sqrt[3]{ab^2}$, $b^3\sqrt[3]{ab}$, $b\sqrt[6]{a^5}$.]

4 Vypočtěte:

1) $\sqrt[4]{16^2}$, $\sqrt[6]{2^{12}}$, $\sqrt[4]{3^8}$, $\sqrt[3]{8\cdot 3^3}$;
2) $(\sqrt{x})^2$, $(\sqrt[3]{x^4})^5$, $(x^3\sqrt[4]{x})^4$, $\sqrt[3]{x}\cdot(\sqrt[3]{x})^3$, $x\geq 0$.

[**1)** 4, 4, 9, 6; **2)** x, $x^6\sqrt[3]{x^2}$, x^{13}, $x\sqrt[3]{x}$.]

5 Vypočtěte:

1) $\sqrt{\frac{25}{16}}$, $\sqrt[3]{\frac{8}{27}}$, $\sqrt[4]{\frac{16}{81}}$;
2) $\frac{\sqrt[3]{56}}{\sqrt[3]{7}}$, $\sqrt{\frac{5}{9}}$, $\sqrt{\frac{81}{7}}$;
3) $\sqrt{\frac{162}{2\,268}}$;
4) $\sqrt{\frac{784}{3\,825}}$.

[**1)** $\frac{5}{4}$, $\frac{2}{3}$, $\frac{2}{3}$; **2)** 2, $\frac{\sqrt{5}}{3}$, $\frac{9}{\sqrt{7}}$; **3)** $\sqrt{\frac{2\cdot 3^4}{2^2\cdot 3^4\cdot 7}}=\sqrt{\frac{1}{14}}$; **4)** $\sqrt{\frac{2^4\cdot 7^2}{3^2\cdot 5^2\cdot 17}}=\frac{2^2\cdot 7}{3\cdot 5}\cdot\frac{1}{\sqrt{17}}$.]

6 Vypočtěte:

1) $\sqrt{\sqrt{2}}$, $\sqrt[3]{\sqrt{3}}$, $\sqrt{\sqrt[3]{5}}$, $\sqrt[3]{\sqrt[3]{3}}$, $\sqrt[3]{\sqrt[3]{2}}$, $\sqrt[3]{\sqrt[4]{27}}$;
2) $\sqrt{\sqrt{x}}$, $\sqrt{\sqrt[3]{x}}$, $\sqrt[3]{\sqrt[4]{x^3}}$, $x\geq 0$.

[**1)** $\sqrt[4]{2}$, $\sqrt[6]{3}$, $\sqrt[6]{5}$, $\sqrt[9]{3}$, $\sqrt[9]{2}$, $\sqrt[4]{3}$; **2)** $\sqrt[4]{x}$, $\sqrt[6]{x}$, $\sqrt[4]{x}$.]

Upravte:

1) $3\sqrt{\frac{5}{3}}$, $10\sqrt{0{,}04}$, $-4\sqrt{\frac{3}{8}}$; 2) $\sqrt[3]{2\sqrt{5}}$, $\sqrt[3]{x\sqrt{x}}$, $x \geq 0$.

[1) $\sqrt{15}$, 2, $-\sqrt{6}$; 2) $\sqrt[6]{20}$, $\sqrt{x}$.]

Vypočtěte:

1) $\sqrt[4]{25}$, $\sqrt[6]{2^{12}}$, $\sqrt[4]{3^8}$, $\sqrt[3]{8 \cdot 3^3}$, $\sqrt[6]{27^4}$;

2) $\sqrt[6]{a^{15}}$, $\sqrt[9]{a^6 b^3}$, $\sqrt{\frac{16}{81}x^4y^8}$, $\sqrt[4]{a^{16}b^8c^4}$.

[1) $\sqrt{5}$,4,9,6,9; 2) $a^2\sqrt{a}$, $\sqrt[3]{a^2b}$, $\frac{2}{3}xy^2$, a^4b^2c.]

Vypočtěte:

1) $\sqrt[4]{(-5)^2}$, $\sqrt{(-7)^2}$, $\sqrt[3]{(-2)^8}$, $\sqrt{m^2}$;
2) $\sqrt{(x+2)^2}$, $\sqrt{x^2+10x+25}$, $\sqrt{9a^2-6\sqrt{2}a+2}$;
3) $\sqrt{\frac{(a+1)^2}{4}}$;
4) $\sqrt{\frac{25x^2-50x+25}{x^2+4x+4}}$;
5) $\sqrt{(x+1)^2(x-3)^2}$;
6) $\sqrt{y^2(4y^2+20y+25)}$.

[1) $\sqrt{5}$, $7{,}4\sqrt[3]{4}$, $|m|$; 2) $|x+2|$, $|x+5|$, $|3a+\sqrt{2}|$; 3) $\frac{|a+1|}{2}$
4) $\frac{|5x-5|}{|x+2|}$, $x \neq -2$; 5) $|x^2-2x-3|$; 6) $|y(2y+5)|$.]

10 Proveďte:

1) $2\sqrt{3}-5\sqrt{4\cdot 3}+\sqrt{9\cdot 3}-\sqrt{25\cdot 3}$;
2) $5\sqrt{6}-7\sqrt{24}+3\sqrt{54}-2\sqrt{96}+2\sqrt{150}$;
3) $10\sqrt[3]{5}-7\sqrt[3]{40}+5\sqrt[3]{135}-4\sqrt[3]{320}+2\sqrt[3]{625}$;
4) $a\sqrt{x}-b\sqrt{x^3}+c\sqrt{x^5}$;
5) $\sqrt{x-y}+\sqrt{16x-16y}+\sqrt{a^2x-a^2y}-\sqrt{9(x-y)}$;
6) $\frac{2}{3}a\sqrt{9a}+6a\sqrt{a}+3\sqrt{a^3}-5a^2\sqrt{\frac{1}{a}+\frac{1}{a^2}\sqrt{a^7}}$.

[1) $-10\sqrt{3}$; 2) $2\sqrt{6}$; 3) $5\sqrt[3]{5}$; 4) $(a-bx+cx^2)\sqrt{x}$, $x \geq 0$;
5) $(2+|a|)\sqrt{x-y}$, $a \in \mathbb{R} \wedge x-y \geq 0$; 6) $7a\sqrt{a}$, $a > 0$.]

11 Proveďte:

1) $\sqrt[3]{6}\cdot\sqrt[3]{16}\cdot\sqrt[3]{18}$;
2) $\sqrt[3]{2x^2y}\cdot\sqrt[3]{3xy^2}\cdot\sqrt[3]{4xy}\cdot\sqrt[3]{9x^2y^2}$;
3) $\sqrt[3]{\frac{2x}{y}}\cdot\sqrt[3]{\frac{y}{3x}}\cdot\sqrt[3]{\frac{4x^2}{y}}\cdot\sqrt[3]{\frac{y^2}{9x}}$;
4) $(2\sqrt{3}-3\sqrt{2}+4\sqrt{6})\cdot\sqrt{6}$;
5) $\sqrt{2}(\sqrt{3}+\sqrt{6})+\sqrt{3}(\sqrt{2}+\sqrt{6})-\sqrt{6}(\sqrt{2}+\sqrt{3})$;
6) $(3\sqrt{2}-2\sqrt{3})(\sqrt{3}+\sqrt{2})$;
7) $(1+\sqrt{2})(\sqrt{5}+3\sqrt{10})$;
8) $(7\sqrt{5}-2\sqrt{3})(3\sqrt{3}+2\sqrt{15})$;
9) $\sqrt{a}(\sqrt{a}+\sqrt{b})-\sqrt{b}(\sqrt{a}-\sqrt{b})$;
10) $(15\sqrt{2}+10\sqrt{3}-6\sqrt{5}-3\sqrt{30})(\sqrt{3}-\sqrt{2})$;
11) $\left(a\sqrt[6]{\frac{b^2}{a}}+b\sqrt[6]{\frac{a^2}{b}}\right)\sqrt[6]{ab}$;
12) $(\sqrt[3]{a^2b}+\sqrt[3]{ab^2})(a\sqrt[3]{ab^2}-ab+b\sqrt[3]{a^2b})$.

[1) 12; 2) $6x^2y^2$, $x \geq 0 \wedge y \geq 0$; 3) $\frac{2}{3}\sqrt[3]{xy}$, $x > 0$, $y > 0$;
4) $6(\sqrt{2}-\sqrt{3}+4)$; 5) $2\sqrt{6}$; 6) $\sqrt{6}$; 7) $7\sqrt{5}+4\sqrt{10}$;
8) $21\sqrt{15}-18+70\sqrt{3}-12\sqrt{5}$; 9) $a+b$, $a \geq 0 \wedge b \geq 0$;
10) $5\sqrt{6}-3\sqrt{10}$; 11) $a\sqrt{b}+b\sqrt{a}$, $a > 0 \wedge b > 0$;
12) a^2b+ab^2, $a \geq 0 \wedge b \geq 0$.]

12 Proveďte:

1) $(5\sqrt{2}+4\sqrt{3})(5\sqrt{2}-4\sqrt{3})$;
2) $(\sqrt{2}+\sqrt{3}+\sqrt{5})(\sqrt{2}+\sqrt{3}-\sqrt{5})$;
3) $(2{,}5\sqrt{0{,}4}-0{,}4\sqrt{2{,}5})^2$;

4) $(\sqrt{4-\sqrt{7}}+\sqrt{4+\sqrt{7}})^2$;
5) $(2\sqrt{a}-3\sqrt{b})(2\sqrt{a}+3\sqrt{b})$;
6) $(3\sqrt{x}-2\sqrt{y})^2$;
7) $(\sqrt{2}+\sqrt{6})^3-(\sqrt{2}-\sqrt{6})^3+(\sqrt{2}+\sqrt{6})^2\cdot\sqrt{3}$;
8) $(3-\sqrt{5})^2+(1+\sqrt{5})^2-(2\sqrt{5}-1)^2$;
9) $(\sqrt{x}+\sqrt{y}+3)^2-(\sqrt{x}+\sqrt{y}-3)^2$;
10) $(\sqrt{x+y}+\sqrt{x}-\sqrt{y})(\sqrt{x+y}-\sqrt{x}+\sqrt{y})$.

[**1)** 2; **2)** $2\sqrt{6}$; **3)** 0,9; **4)** 14; **5)** $4a-9b$, $a\geq 0\ \wedge\ b\geq 0$; **6)** $9x-12\sqrt{xy}+4y$, $x\geq 0\ \wedge\ y\geq 0$; **7)** $24\sqrt{6}+8\sqrt{3}+1$ **8)** -1; **9)** $12(\sqrt{x}+\sqrt{y})$, $x\geq 0\ \wedge\ y\geq 0$; **10)** $2\sqrt{xy}$, $x\geq 0\ \wedge\ y\geq 0$.]

13 Proveďte:

1) $\sqrt[3]{a^2}\cdot\sqrt[6]{a^5}$;
2) $\sqrt[4]{m^3}\cdot\sqrt[6]{m}\cdot\sqrt[8]{m^5}$;
3) $\left(2\sqrt[3]{3}-3\sqrt{2}\right)\sqrt[6]{6}$;
4) $\sqrt{2ab}\cdot\sqrt[3]{4a^2b^4}\cdot\sqrt[4]{8a^3b^5}\cdot\sqrt[6]{a^5b^7}\cdot\sqrt[12]{2a^3b^9}$;
5) $\sqrt[6]{a^5}:\sqrt[9]{a^4}$;
6) $\sqrt[6]{a^9x^5}:\sqrt[4]{a^2x^3}$;
7) $\sqrt{\dfrac{5\sqrt[3]{3}}{6}}:\sqrt[3]{\dfrac{6\sqrt{5}}{3\sqrt{3}}}$.

[**1)** $a\sqrt{a}$; **2)** $m\sqrt[24]{m^{13}}$; **3)** $2\sqrt[6]{54}-3\sqrt[6]{48}$; **4)** $4a^3b^5$; **5)** $\sqrt[18]{a^7}$; **6)** $a\sqrt[12]{x}$; **7)** $\dfrac{\sqrt[18]{3}}{\sqrt{2}}$.]

14 Odstraňte odmocniny ze jmenovatele zlomku (usměrněte):

1) $\dfrac{1}{\sqrt{2}}$;
2) $\sqrt{\dfrac{5}{3}}$;
3) $\dfrac{3\sqrt{2}}{4\sqrt{5}}$;
4) $\dfrac{\sqrt{5}+\sqrt{2}}{\sqrt{2}}$;
5) $\dfrac{6+\sqrt{12}}{\sqrt{3}}$;
6) $\dfrac{2-\sqrt{3}}{\sqrt{5}}$;
7) $\dfrac{\left(\sqrt{2}+1\right)^3+\left(\sqrt{2}-1\right)^3}{2\sqrt{2}}$;
8) $\dfrac{a}{\sqrt{b}}$, $b>0$;
9) $\sqrt{\dfrac{m}{n}}$, $m\geq 0\ \wedge\ n>0$;
10) $\dfrac{a+\sqrt{b}}{\sqrt{b}}$, $b>0$.

[**1)** $\dfrac{\sqrt{2}}{2}$; **2)** $\dfrac{\sqrt{15}}{3}$; **3)** $\dfrac{3\sqrt{10}}{20}$; **4)** $\dfrac{2+\sqrt{10}}{2}$; **5)** $2\sqrt{3}+2$; **6)** $\dfrac{2\sqrt{5}-\sqrt{15}}{5}$; **7)** 5; **8)** $\dfrac{a\sqrt{b}}{b}$; **9)** $\dfrac{\sqrt{mn}}{n}$; **10)** $\dfrac{a\sqrt{b}+b}{b}$.]

15 Usměrněte:

1) $\dfrac{1}{2+\sqrt{3}}$;
2) $\dfrac{3}{\sqrt{3}+\sqrt{2}}$;
3) $\dfrac{4}{3-\sqrt{2}}$;
4) $\dfrac{19\sqrt{6}}{9-2\sqrt{6}}$;
5) $\dfrac{5-\sqrt{3}}{5+\sqrt{3}}$;
6) $\dfrac{a}{1-\sqrt{a}}$, $a\geq 0\ \wedge\ a\neq 1$;
7) $\dfrac{19\sqrt{6}}{5\sqrt{3}+3\sqrt{2}}$;
8) $\dfrac{a}{\sqrt{a}+\sqrt{b}}$, $a>0\ \wedge\ b>0\ \wedge\ a\neq b$.

[**1)** $2-\sqrt{3}$; **2)** $3(\sqrt{3}-\sqrt{2})$; **3)** $\dfrac{4(3+\sqrt{2})}{7}$; **4)** $3\sqrt{6}+4$; **5)** $\dfrac{14-5\sqrt{3}}{11}$; **6)** $\dfrac{a(1+\sqrt{a})}{1-a}$; **7)** $5\sqrt{2}-2\sqrt{3}$; **8)** $\dfrac{a(\sqrt{a}-\sqrt{b})}{a-b}$.]

16 Proveďte:

1) $1+\dfrac{1+\sqrt{3}}{2+\sqrt{3}}$;
2) $\dfrac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}+\dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$;
3) $\dfrac{2}{\sqrt{3}-1}+\dfrac{3}{\sqrt{3}-2}+\dfrac{15}{3-\sqrt{3}}$;
4) $\dfrac{a}{\sqrt{3}-1}+\dfrac{a}{\sqrt{3}+1}$;
5) $\dfrac{5\sqrt{2}}{3-\sqrt{3}}+\dfrac{\sqrt{2}}{3+\sqrt{3}}$;
6) $\dfrac{a+\sqrt{b}}{a-\sqrt{b}}-\dfrac{a-\sqrt{b}}{a+\sqrt{b}}$;
7) $\dfrac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}+\dfrac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}}$;

8) $\dfrac{1-\sqrt{x}}{1+\sqrt{x}}+\dfrac{3\sqrt{x}}{1-\sqrt{x}}-\dfrac{3+\sqrt{x}}{1-x}$;

9) $\dfrac{\sqrt{6}}{2\cdot\sqrt{2}-\sqrt{5}}-\dfrac{\sqrt{10}+3}{\sqrt{3}}$;

10) $\dfrac{\dfrac{\sqrt{2}-1}{2}-\dfrac{2}{\sqrt{2}+1}}{\sqrt{2}-1+\dfrac{2}{\sqrt{2}+1}}$.

[**1)** $\sqrt{3}$; **2)** 10; **3)** $\dfrac{5+\sqrt{3}}{2}$; **4)** $a\sqrt{3}$; **5)** $\dfrac{9\sqrt{2}+2\sqrt{6}}{3}$;
6) $\dfrac{4a\sqrt{b}}{a^2-b}$; **7)** $\dfrac{2(a+b)}{a-b}$; **8)** $\dfrac{4x-2}{1-x}$; **9)** $\dfrac{\sqrt{3}}{3}$; **10)** $-\dfrac{1}{2}$.]

4.2.17 Usměrněte:

1) $\dfrac{20}{\sqrt{7}-\sqrt{5}+\sqrt{2}}$;

2) $\dfrac{2}{\sqrt[3]{3}}$;

3) $\dfrac{4}{\sqrt[3]{5}-1}$;

4) $\dfrac{7}{\sqrt[3]{5}+\sqrt[3]{2}}$;

5) $\dfrac{3}{\sqrt[3]{25}+\sqrt[3]{10}+\sqrt[3]{4}}$;

6) $\dfrac{12}{\sqrt{15}-\sqrt{6}+\sqrt{35}-\sqrt{14}}$;

7) $\dfrac{1}{2+\sqrt[3]{2}+\sqrt[3]{4}}$;

8) $\dfrac{1}{\sqrt{a+\sqrt{a^2-b^2}}}$, $a \geq b > 0$.

[**1)** $\sqrt{70}-2\sqrt{5}+5\sqrt{2}$; **2)** $\dfrac{2\sqrt[3]{9}}{3}$; **3)** $\sqrt[3]{25}+\sqrt[3]{5}+1$;
4) $\sqrt[3]{25}-\sqrt[3]{10}+\sqrt[3]{4}$; **5)** $\sqrt[3]{5}-\sqrt[3]{2}$; **6)** $(\sqrt{5}+\sqrt{2})(\sqrt{7}-\sqrt{3}$
7) $1-\dfrac{1}{2}\sqrt[3]{4}$; **8)** $\dfrac{\sqrt{a-\sqrt{a^2-b^2}}}{b}$.]

4.2.18 Upravte výrazy:

1) $\left(\dfrac{\sqrt{1+a}}{\sqrt{1+a}-\sqrt{1-a}}+\dfrac{1-a}{\sqrt{1-a^2}-1+a}\right)\left(\sqrt{\dfrac{1}{a^2}-1}-\dfrac{1}{a}\right)$
$0<a<1$;

2) $\left(\dfrac{\sqrt[4]{a^3}-\sqrt[4]{b^3}}{\sqrt{a}-\sqrt{b}}-\sqrt[4]{a}-\sqrt[4]{b}\right)\left(\sqrt[4]{\dfrac{a}{b}}+1\right)$;

3) $\left[\dfrac{1-a^2}{\left(\dfrac{1-a\sqrt{a}}{1-\sqrt{a}}+\sqrt{a}\right)\left(\dfrac{1+a\sqrt{a}}{1+\sqrt{a}}-\sqrt{a}\right)}+1\right]\cdot\sqrt{1-a^2}$;

4) $\dfrac{\sqrt{a}-2\sqrt{a}+1}{\sqrt{a}-2\sqrt[4]{a}+1}:\dfrac{\sqrt[4]{a}+1}{\sqrt[4]{a}-1}+1$;

5) $\left(\sqrt{\dfrac{1}{1-m^2}}-\sqrt{1-m^2}\right):$
$:\left(\dfrac{1}{4+4\sqrt{m}}+\dfrac{1}{4-4\sqrt{m}}-\dfrac{1}{2+2m}\right)$, $0<m<1$;

6) $\left(\dfrac{a+1}{\sqrt{a}}+\dfrac{1}{a-\sqrt{a}}-\dfrac{a}{\sqrt{a}+1}\right)\dfrac{\sqrt{3}-a\sqrt{3}}{a+1}$, $a>1$;

7) $\left(\sqrt{a(1-a)}+\dfrac{\sqrt{a^3}}{\sqrt{1-a}}\right):\left(\dfrac{1}{1+\sqrt{a}}+\dfrac{\sqrt{a}}{1-a}\right)$, $0<a<1$.

[**1)** -1; **2)** $-\sqrt[4]{a}$; **3)** 2 pro $0<a<1$, -2 pro $a>1$;
4) 0 pro $0<a<1$, 2 pro $a>1$; **5)** $m\sqrt{1-m^2}$; **6)** $-\sqrt{3}$;
7) $\sqrt{a(1-a)}$.]